Skills Summary

Counting Steps, the Line Behind a Sequence, and Counting Terms

Use this reference while completing your worksheet or when studying.

Skill 1 — Count steps, not terms

Explicit rule: $a_n = a_1 + (n-1)d$, where d is the common difference.

- The first term costs no steps, so reaching term n takes $n-1$ of them.
- **The $n = 1$ test:** substitute $n = 1$. A correct rule returns a_1 exactly; a rule using nd returns the second term instead.
- Going backwards works too: to find *which* term equals a value V, solve $a_1 + (n-1)d = V$, then substitute your n back in.

Example: An arithmetic sequence has $a_1 = 4$ and $d = 9$. Find the 11th term.
Step 1. The explicit rule jumps straight to any term, and its whole risk is the step count, so write $(n-1)$ before substituting a single number.
Step 2. $a_{11} = 4 + (11-1)(9) = 4 + 90$
The 11th term is **94**.

Try it: An arithmetic sequence has $a_1 = 7$ and $d = 5$. Find the 13th term.

check: **67**

Skill 2 — The line behind the sequence

Linear form: $a_n = dn + b$ with $b = a_1 - d$.

- The slope is the common difference d: one step along, one d up.
- The constant b is the value at $n = 0$ — one step *before* the sequence starts. It is the y-intercept of the line, never a_1.
- Same $n = 1$ test: $d(1) + (a_1 - d) = a_1$ ✓. If your rule returns a_2 at $n = 1$, you used a_1 as the constant.

Example: The sequence 5, 12, 19, … has $a_1 = 5$ and $d = 7$. Write it in linear form and find the 9th term.
Step 1. The slope is the common difference, and the constant is one step below the first term, so build $b = a_1 - d$ before touching the index.
Step 2. $b = 5 - 7 = -2$, so $a_n = 7n - 2$ and $a_9 = 7(9) - 2 = 63 - 2$
The 9th term is **61**.

Try it: The sequence 8, 13, 18, … has $a_1 = 8$ and $d = 5$. Write it in linear form and find the 12th term.

check: **63**

Skill 3 — How many terms are in the sum?

Term count: from term p through term q inclusive there are $q - p + 1$ terms. **Sum:** $S = \dfrac{(\text{number of terms})}{2}(\text{first} + \text{last})$.

- Subtracting index numbers counts *gaps*; adding 1 turns gaps back into terms.
- The last term of the first n terms is $a_1 + (n-1)d$ — using $a_1 + nd$ sums one term too many.
- Doubtful? Test your counting rule on a three-term stretch you can count on your fingers.

Example: Find the sum of the first 10 terms of 2, 7, 12, ...

Step 1. The sum formula needs the true last term, so find a_{10} with the $(n-1)$ rule first rather than reaching for $a_1 + nd$.

Step 2. $a_{10} = 2 + 9(5) = 47$, so $S_{10} = \dfrac{10}{2}(2 + 47) = 5 \times 49$

The sum is **245**.

Try it: Find the sum of the first 8 terms of 3, 10, 17, ...

check: **220**

(!) Watch out: An off-by-one index almost always produces a clean, whole number, so nothing looks wrong. Run the $n = 1$ test on every rule you write, and substitute your answer back into the sequence before you trust it.